run_sjsdm_predictor_structure_folds.R

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Contents

`run_sjsdm_predictor_structure_folds` *Run One sjSDM Predictor Structure Across Selected Folds*

Description

Applies one controlled predictor structure to deterministic selected-candidate folds and returns labelled predictions and diagnostics.

Usage

```
run_sjsdm_predictor_structure_folds(
  predictor_structure = NULL,
  data_assignments = NULL,
  data_selected_candidate = NULL,
  data_community_matrix = NULL,
  data_abiotic_wide = NULL,
  data_coords_projected = NULL,
  data_sample_ids = NULL,
  config_model_fitting = NULL,
  config_data_processing = NULL,
  model_formula = NULL,
  device = "gpu",
  seed = 900723L,
  prepare_fold_function = prepare_sjsdm_cross_validation_fold,
  fit_candidate_function = fit_sjsdm_regularization_candidate,
  runner_function = run_sjsdm_selected_candidate_folds,
  predict_function = predict_sjsdm_probability_matrix
)
```

Arguments

predictor_structure
: Structure passed to [configure_sjsdm_predictor_structure()].

data_assignments, data_selected_candidate, data_community_matrix
: Cross-validation assignments, selected regularization, and response matrix.

data_abiotic_wide, data_coords_projected, data_sample_ids
: Abiotic predictors, projected coordinates, and aligned sample metadata.

config_model_fitting, config_data_processing
: Model-fitting and data-processing configurations used by fold preparation.

model_formula
: Full abiotic formula used when abiotic predictors are enabled.

device
: Fitting device passed to [fit_sjsdm_regularization_candidate()].

seed
: Base selected-fold seed passed to [run_sjsdm_selected_candidate_folds()].

prepare_fold_function, fit_candidate_function, runner_function
: Injectable functions for fold preparation, candidate fitting, and selected fold orchestration.

predict_function
: Injectable held-out prediction function.

Details

Regularization and assignments are fixed by the supplied reference inputs. The helper changes predictor inclusion only. Intercept-only and spatial-only structures use an environmental intercept because sjSDM requires an environmental component. The no-associations structure retains abiotic and spatial predictors while restricting the biotic covariance to its diagonal.

Value

Named list containing labelled ‘data_predictions‘ and ‘data_diagnostics‘.

Examples

```
## Not run:
run_sjsdm_predictor_structure_folds(
  predictor_structure = "abiotic_only",
  data_assignments = data_assignments,
  data_selected_candidate = data_selected_candidate,
  data_community_matrix = data_community_matrix,
  data_abiotic_wide = data_abiotic_wide,
  data_coords_projected = data_coords_projected,
  data_sample_ids = data_sample_ids,
  config_model_fitting = config_model_fitting,
  config_data_processing = config_data_processing,
  model_formula = model_formula
)

## End(Not run)
```

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